

Sandboxing and resource limits

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Untrusted input boundaries

Without opening the old lesson, explain one key idea from Untrusted input boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Sandboxing and resource limits. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Sandboxing and resource limits. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Queue currently has 9 tasks; job proposes 2 more under a maximum10 policy. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

A job requests a public webpage and a write outside its output directory. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Resource limits constrain execution but cannot prove generated claims or algorithms correct. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Queue currently has 9 tasks; job proposes 2 more under a maximum10 policy. Decide.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled sandboxing and resource limits evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of sandboxing and resource limits. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Sandboxing and resource limits

Bound a student program. A toy sandbox forbids network access, allows at most two seconds and limits output to 20 MB.

A uses one second and writes 2 MB. B tries to contact a website. These are declared trial records.

Sandboxing and resource limits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply isolation

Neither program receives network permission.



2. Check A

One second and 2 MB fit both resource limits.



3. Deny B network call

Stop the prohibited network action regardless of its purpose.

CHANGE ONE THING

A takes three seconds but still writes 2 MB.

Explain the changed result using the matching lesson.

Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its output remains only 2 MB?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Bound a student program. A toy sandbox forbids network access, allows at most two seconds and limits output to 20 MB.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Stop the prohibited network action regardless of its purpose.

1. Deny B network call
2. Apply isolation
3. Check A

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Deny B network call Stop the prohibited network action regardless of its purpose. Sandbox controls bound reach and resource use. Passing an output limit cannot cancel a network or time restriction.

Practice 2: explain the deciding evidence

Changed-case practice: A takes three seconds but still writes 2 MB. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Stop A for exceeding the time allowance.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Stop A for exceeding the time allowance. The independent two-second limit remains binding.

Repair a missing building block

Review Untrusted input boundaries (ai-6/18). Goal: Explain the mechanism and terminology of untrusted input boundaries. Short practice: Classify h3, H3, h30 and the trailing-space input by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the mechanism and terminology of timeouts retries and cancellation. Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?. Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram. Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
